2D Design

Best Practice

A technical drawing is made in universal language using norms, standards and internal agreements. The recommendation is to keep textual data to a minimum.

The drawings must always be aligned and contain the correct information.

General remarks-

Manufacturability of part and machining.

Clear and organised drawings.

Part numbering in an assembly.

Standards and correct views on drawings.

PED data.

Marking and Dimensioning-

Group dimensions.

General tolerances.

Welding indications.

Marking parts and threads according to AC standards.

Indicate details in ISO view.

Drawing Notes.

Drawing Rules

1. Formats

Drawing sheets according *"en 1220 K"*.

SIZE OF DRAWING SHEETS

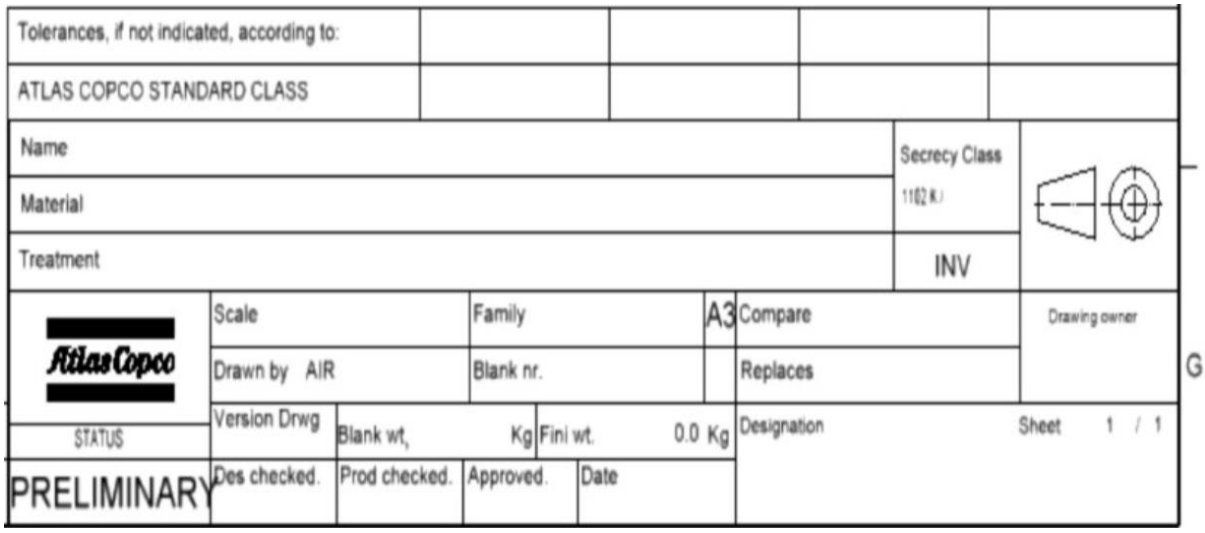
The original drawing should be made on the smallest sheet permitting the necessary clarity and unambiguity.
If space so requires the drawing may be made on more than one sheet.
Drawing sheets should be chosen from the basic series ISO-A according to ISO 5457:1999 as shown in table 1.

(Size A0 should be avoided)

Standard sizes

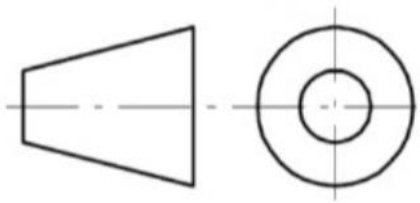
Designation	Dimensions (mm)	Layout
(A0)	841 × 1189	horizontal
A1	594 × 841	horizontal
A2	420 × 594	horizontal
A3	297 × 420	horizontal
A4	210 × 297	vertical

Title Block



Projection symbol

Although the first angle projection method (method E) applies as Atlas Copco standard, the projection method should always be indicated on the drawing. Certain incorporated companies use method A; therefore, indication is essential. Symbol shall follow ISO 5456-2:1996 Annex A.



.Symbol for first angle projection method (Method E).

Projection Methods

Three methods for projection exist in international standard:

- The third angle projection method (arrow shown)
- The reference arrow method (formerly denoted as method E)
- The first angle projection method (formerly denoted as method A)

Difference between reference arrow and first angle method is shown in the drawings below.

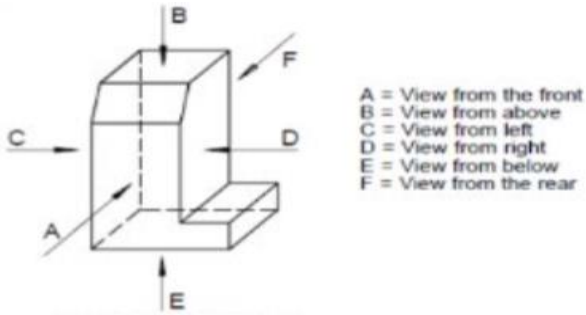


Figure 1.Different views.

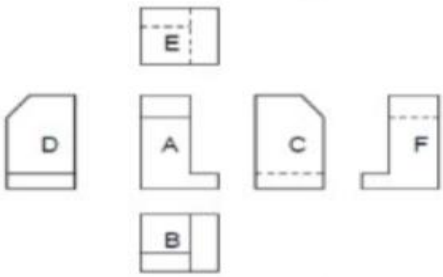


Figure 2.First angle projection method (Method E), recommended within Atlas Copco.

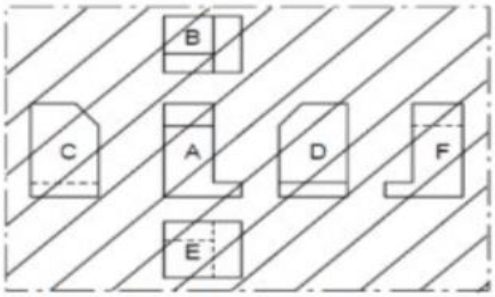


Figure 3.Third angle projection method (Method A).

Secrecy class

Class 1:
Documents about products or production methods containing essential product info, development project details, or unique features.

Class 2:
Documents containing other important product information.

Class 3:
Documents specifying trivial articles without significant influence on product features.

Responsibility for assigning classes = design department manager.

Secrecy class (1–3) must be clearly stated on each document.
See AC-STD 1220 K for drawing secrecy class placement.

Scales

Scales according *“en 1210 K”*.

The table below shows recommended drawing scales.
Whenever possible, drawings should be prepared in scale **1:1**.

Drawing scale table

Full size	Reduced view	Reduced view	Reduced view
1:1	1:2	1:5	1:10
	1:20	1:50	1:100

Enlarged view
2:1
5:1
10:1

Designation / Numbering of drawings

GENERAL

Rules and guidelines for numbering articles within Atlas Copco Group.

Each article must have a **unique article number**, enabling clear distinction.

Atlas Copco article numbering ensures rational and reliable handling across companies and suppliers.

Numbering of components

ARTICLE NUMBER BUILD-UP

- Article numbers contain **10 digits**.
- Numbers are normally taken in consecutive order.
- Write article number as groups:1234 5678 90
- Zeros must be written if at beginning/end.

ALLOCATION OF ARTICLE NUMBER SERIES

Companies must request number series from Group Standards department.

Existing numbers should be reused before requesting new series.

<u>Specific numbering rules for Industrial Air</u>

1. Use **.00** for a part + all non-dismountable components
   (.02 / .03 / .04 ... used for non-dismountable components)
2. Use **.80** for assemblies of mountable parts
   (decals, foam, locks, etc.)
3. Additional number ranges:
   - **.40**: Multi-brand colours by plates
   - **.50**: Multi-brand colours by plates
   - **.60**: Panel sets (older)
   - **.70**: As-cast drawing
   - **.90**: Marine assembly drawings

Table drawings:

- Use **.00** drawing number
- Numbering is consecutive; above rules do NOT apply

Detailed Engineering Drawing

[Diagram only — engineering drawing sheet, no transcribable text]

4) Vessel assembly

- Table drawing is a **.80-drawing**
- All variants defined on this .80-drawing

Dedicated extension numbers

Extension	Approval
82	Lloyds / AS1210
83	Lloyds / Bureau Veritas
85	MHLV
86	ML
95	MOM
96	TÜV

Extension	Approval
99	CE / ASME

Please consult QA-instruction T-04.11
“Instruction for drawings of welded pressure vessels”

6c. Title block

- No material indication for standard parts
- No material indication on **.80** drawings → use *“Not Applicable”*
- Remove *Material* + *Description* from part list
- No edition indication of parts on **.80** drawings
- Fill “Compare” and “Replaces” when possible
- Plate/material thickness always added in part list/title block; never dimensioned on part itself

6b. Numbering of components

General rule: add the part number on the part itself
Exception: metal sheet components (treatment may remove marking)

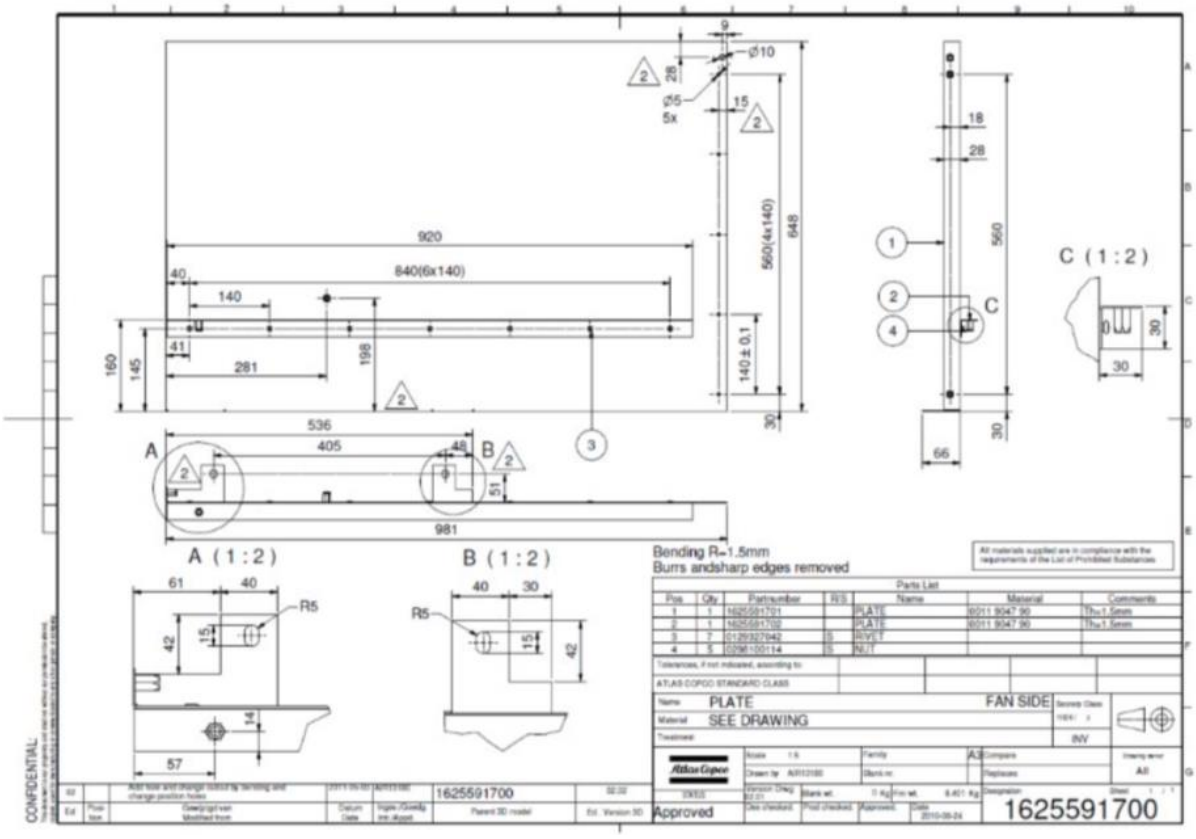
Supplier references allowed **only for A-components**
If insufficient specification, metadata must distinguish similar parts; otherwise, qualification program required.

Foam rules:

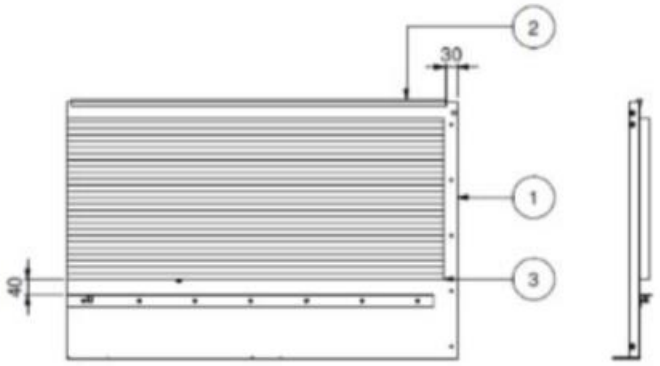
- Always drawn with **adhesive side on top**
- Add “adhesive side” text
- Use **Silencing foam** (Level 3 classification)
- Do not use Level 4 classifications
- Applies to new & updated drawings

8. Hints

- Reserve always the **.00-number**
- No unfolding on the drawing
- 3D view may be added for complex parts


920
840(6x140)
40
140
41
281
160
145
198
536
405
48
51
981
A
B
2
3
9
28
15
Ø10
Ø5
5x
560(4x140)
648
140±0,1
30
18
28
560
66
30
1
2
4
C
C (1:2)
30
30
A (1:2)
61
40
R5
42
15
14
57
B (1:2)
40
30
R5
15
42
Bending R=1.5mm
Burns andsharp edges removed
All materials supplied are in compliance with the requirements of the List of Prohibited Substances
Parts List
Pos. Qty. Partnumber RS Name Material Comments
1 1 1625591701 PLATE 6011 9047 90 Thx1.5mm
2 1 1625591702 PLATE 6011 9047 90 Thx1.5mm
3 7 0129327042 S RIVET
4 5 02M100114 S NUT
Tolerances, if not indicated, according to:
ATLAS COPCO STANDARD CLASS
Name PLATE
FAN SIDE
Material SEE DRAWING
Treatment
INV
Scale 1:5
Family
A3
Compare
Drawn by AR10100
Blank nr.
Replaces
All
Approved
Blank wt. 1 Kg
Fin wt. 0.401 Kg
Date 2010-09-24
1625591700
1625591700
Parent 3D model
Est. Version 3D
CONFIDENTIAL

Parent 3D model:	1625591780	Ed.Vers.3D	02.02
Note / Position	Modified from	Date	Init./Appd.
02	Change plate and snew foam	2011-06-03	AIR13180



All materials supplied are in compliance with the requirements of the List of Prohibited Substances

Parts List

Pos	Qty	Partnumber	R/S	Name	Comments
1	1	1625591700	R	SUPPORT	
2	1	1202658400		SEAL	L=9.4dm
3	1	1625591760	R	INSULATION	

Tolerances, if not indicated, according to:
ATLAS COPCO STANDARD CLASS

Name: SUPPORT — FAN SIDE
Material: SEE DRAWING
Treatment:
Secrecy class: INV

Atlas Copco

Scale: 1:10 | Family: | A4 | Compare: | Drawing name: All
Drawn by: AIR13180 | Blank nr.: | Replaces:
STATE: Approved | Version Dwg: 02.01 | Blank wt.: 0 Kg | Finish wt.: 8.64 Kg | Designation: 1625591780 | Sheet: 1 / 1
Des checked: | Prod checked: | Approved: | Date: 2010-07-30

CONFIDENTIAL:

Line types

Two widths used: wide & narrow (ratio 2:1)

Table — Line types

Line type	Description	Width (mm)
Continuous wide	———————————	0.5–0.7
Continuous narrow	———————————	0.25–0.35
Continuous narrow freehand		0.25–0.35
Dashed narrow	— — — — — — —	0.25–0.35
Long-dashed dotted wide	— - — - — - — - —	0.5–0.7
Long-dashed dotted narrow/wide	Not shown in ISO 128-24. Used for emphasis.	0.25–0.35 / 0.5–0.7
Long-dashed dotted narrow	— - — - — - — - —	0.25–0.35
Long-dashed double-dotted narrow	— -- — -- — -- — - —	0.25–0.35

Dimensioning

<u>Application</u>

- All info necessary to define an article must be placed directly on the drawing unless stated otherwise.
- Each feature dimensioned once only.
- Dimensions placed in the most descriptive view.
- Dimensions in mm; omit “mm”.
- Other units allowed if required (Nm, bar, µm).
- No more dimensions than necessary.
- Auxiliary dimensions permitted when needed for:
    - production stages
    - clarity, etc.

The following symbols are used with dimensions to show applicable shape identification and to improve drawing interpretation.

The applicable symbol should precede the value for the dimension.

Symbol	Meaning
Ø	Diameter
R	Radius
□	Square
SØ	Spherical diameter
SR	Spherical radius

Tolerances

Tolerance Class	Description	≤3	>3–6	>6–30	>30–120	>120–400	>400–1000	>1000–2000	>2000–4000	>4000–8000	>8000–12000	>12000–16000	>16000–20000
f	fine	±0.05	±0.05	±0.1	±0.15	±0.2	±0.3	±0.5	±0.8	±1.2	±2	±3	±4
m	medium	±0.1	±0.1	±0.2	±0.3	±0.5	±0.8	±1.2	±2	±3	±4	±5	±6
c	coarse	±0.2	±0.3	±0.5	±0.8	±1.2	±2	±3	±4	±6	±8	±10	±12
v	very coarse	±0.5	±1	±1.5	±2.5	±4	±6	±8	±10	±12	±16	±20	±?

General Tolerances according "en 1350 K"

<u>Indication on drawings</u>:
Reference to AC-STD 1350 K shall be made in the title block.
General references always apply.

Designation (1350 K) may be stated with or without tolerance classes, but ISO classes must be indicated for applicability.

Note:
ISO tolerance classes shown under:

- paragraph 2.3 (linear/angular)
- paragraph 2.4 (geometrical)

<u>Indirectly stated Tolerances according “en 1350:01 K”</u>

This standard is used for sheet metal (< 8 mm) when ATLAS COPCO 1350 K cannot be applied. For > 8 mm thickness, welding & flame cutting tolerances apply.

Basic dimension	Allowed deviation ±
– 315	1
(315) – 1000	2
(1000) – 2000	3
(2000) – 4000	4
(4000) – …	5

Tolerances of Form and Position

Tolerances	Characteristics	Symbol	Datum need	Subclause
Form	Straightness	⏤	no	10.2.1
	Flatness	▱	no	10.2.2
	Roundness	○	no	10.2.3
	Cylindricity	◯	no	10.2.4
	Profile any line	∩	no	10.2.5
	Profile any surface	⌓	no	10.2.6
Orientation	Parallelism	∥	yes	10.3.1
	Perpendicularity	⊥	yes	10.3.2
	Angularity	∠	yes	10.3.3
Location	Position	⌖	yes	10.4.1
	Concentricity (centre points)	○→○	yes	10.4.2
	Symmetry	≡	yes	10.4.3
Run out	Total run-out	↗↗	yes	10.5.2
	Run out	↗	yes	10.5.1

Additional symbols

Description	Symbol	Reference
Theoretical feature indication	☐ dashed	Clause 7
Datum feature row indication	⌖ row symbol	Clause 10.4 ISO 5458
Datum target indication	◉ target	ISO 5458
Theoretically exact dimension	⌂	Clause 11
Envelope requirement	⏢	Clause 12 (ISO 8015)
Maximum material requirement	Ⓜ (M)	Clause 14 ISO 2692
Least material requirement	Ⓛ (L)	Clause 14 ISO 2692
Free state condition	F	Clause 16 (ISO 8015)
ISP mould profile	ISP	Subdivision 13.2
Common zone	CZ	Subdivision 15
Surface texture requirement	...	Subdivision 19
Local diameter	LD	Subdivision 15.2
Maximum diameter	HP	Subdivision 16.2
Radius	R	Subdivision 18.1
Spherical radius	SR	Subdivision 18.4
Any cross section	ACS	Subdivision 13.11

Tolerance frame

Requirements

Shown in a rectangular frame with compartments arranged left → right:

1. Symbol of geometrical characteristic
2. Tolerance value (preceded by ⌀ if zone is circular)
3. Datum or common datums (letters)

Figure 1

Figure 2

Figure 3

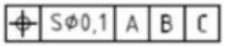


Figure 4

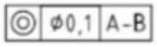


Figure 5

Surface roughness

Surface roughness according **“en 1254 K”**

GENERAL

- Surface roughness should be stated as the arithmetical mean deviation **Ra**.
- Sometimes max height of profile **Rz** may be used.
- Ra should be less than 1/12 of dimensional tolerance.
- Chamfers, keyways, countersinks, etc. indicated only if needed.
- Surface roughness only stated when required for operation/appearance.

Standard values for Ra

Ra values should be selected from:

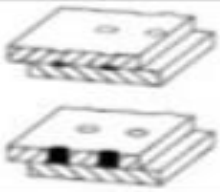
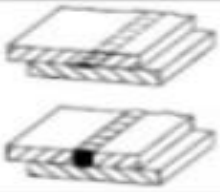
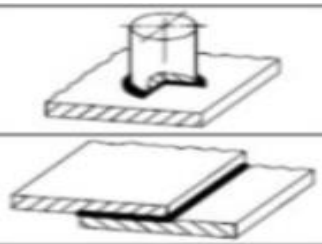
0.1 | 0.2 | 0.4 | 0.8 | 1.6 | 3.2 | 6.3 | 12.5 | 25 | 50

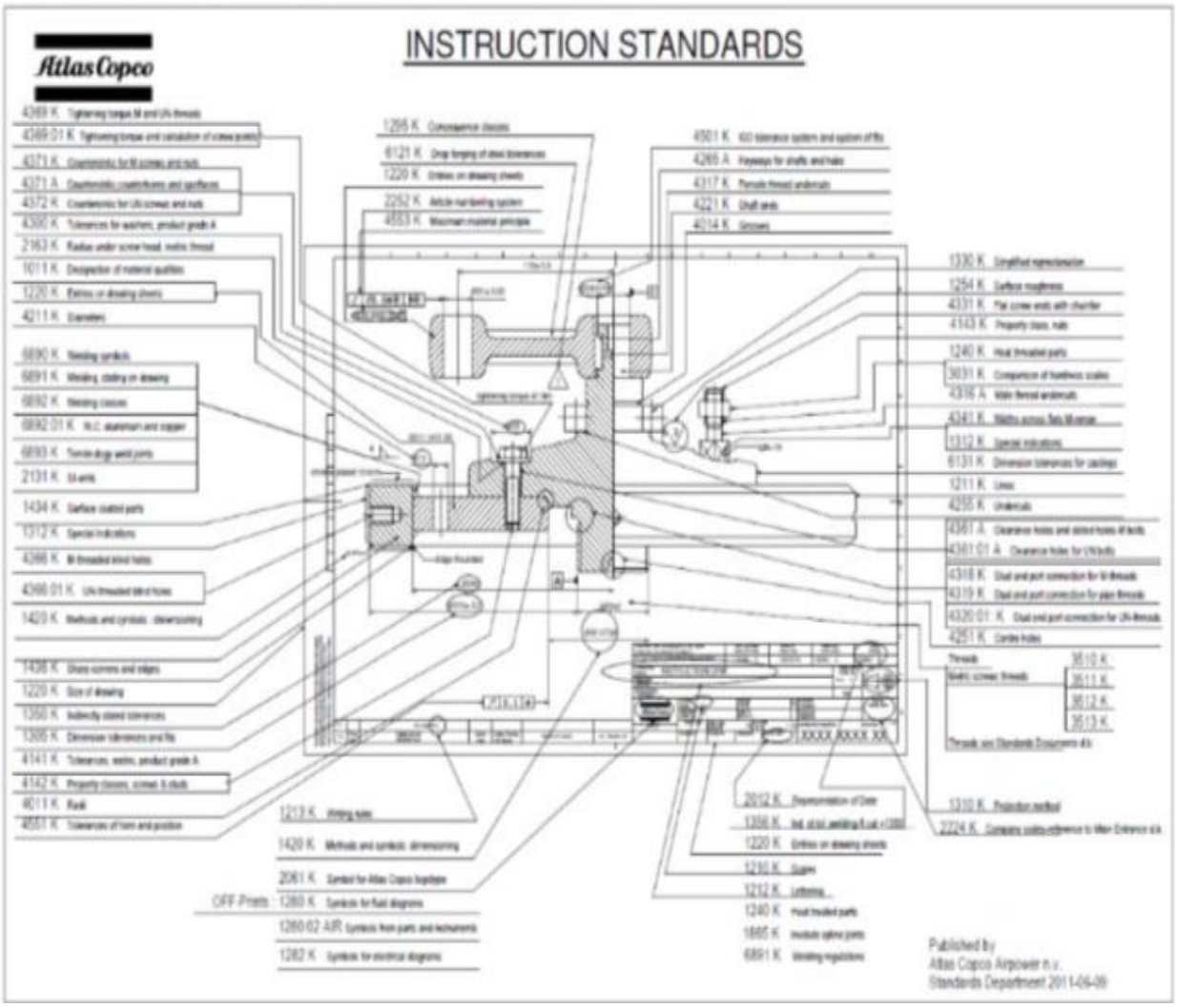
Other values may be used if necessary (e.g., for supplier parts).

SYMBOL	DESCRIPTION
No indication	Unmachined surface obtained by casting, forging, rolling drawing, pressing, gas cutting, etc. Burrs shall be removed.
	Surface obtained with or without removal of material.
	Surface obtained without removal of material. On an operation drawing the symbol indicates that the surface is to be left in the state it has from the previous operation, irrespective of whether this state was achieved by removal of material or not. In this case no R_a value is given.
	Surface obtained by removal of material.
	As the previous symbol but with the additional note that the surface should have been otbtained through machining (in this case grinding).
	As the previous symbol but with an additional symbol indicating that there should be a specific direction of lay (in this case perpendicular to the plane of projection).
	As the previous symbol but with the additional note that point deivations may occur [in this case up to 50 µm (R_y)].

Welding

Table 1. Elementary symbols — Cont.

No.	Designation	Illustration	Symbol
11	Plug weld or slot weld		⊓
12	Spot weld		○
13	Seam weld		⊖
14	Steep-flanked single-V butt weld		V
15	Steep-flanked single-bevel butt weld		V
16	Edge weld		|||
17	Surfacing		⌒
18	Surface joint		=
19	Inclined joint		⫽
20	Fold joint		⫏


INSTRUCTION STANDARDS
Atlas Copco
4369 K.
4369-01 K.
4371 K.
4371 A.
4372 K.
4300 K.
2163 K.
1011 K.
1220 K.
4211 K.
6890 K.
6891 K.
6892 K.
6892-01 K.
6893 K.
2131 K.
1434 K.
1312 K.
4366 K.
4366-01 K.
1420 K.
1408 K.
1220 K.
1350 K.
1305 K.
4141 K.
4142 K.
4011 K.
4551 K.
1295 K.
6121 K.
1220 K.
2252 K.
4553 K.
4501 K.
4265 A.
4317 K.
4221 K.
4014 K.
1330 K.
1254 K.
4331 K.
4143 K.
1240 K.
3031 K.
4316 A.
4341 K.
1312 K.
6131 K.
1211 K.
4255 K.
4361 A.
4361-01 A.
4318 K.
4319 K.
4320-01 K.
4251 K.
Threads
3610 K.
3611 K.
3612 K.
3613 K.
1213 K.
1420 K.
2061 K.
OFF-Prints
1260 K.
1260-02 AIR
1282 K.
2012 K.
1356 K.
1220 K.
1230 K.
1212 K.
1240 K.
1665 K.
6891 K.
1310 K.
2224 K.
Published by
Atlas Copco Airpower n.v.
Standards Department 2011-06-09